



Unintended Consequences of Disclosing Recommendations by Artificial Intelligence versus Humans on True and Fake News Believability and Engagement

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ABSTRACT

In an attempt to combat fake news, policymakers in many countries are considering mandating the disclosure of artificial intelligence (AI) recommendations of social media news articles. We used two randomized controlled experiments to investigate the effects of labeling social media news stories as recommended by AI. Our results show that an AI recommendation reduced belief in true news articles and had no material effect on belief in fake news. In contrast, a recommendation by an expert increased belief in true news articles, but had no effect for fake news articles. A friend recommendation had no effect for fake articles and inconsistent effects for true articles. Belief that an article was true led to news engagement (liking, commenting, sharing), but an AI recommendation weakened this relationship, making confirmation bias the primary factor influencing engagement. The trustworthiness of the recommender only partially explained these effects, which suggests that there are other theoretical factors at work. This study reveals that the explicit labeling of AI curation of social media news stories does not help combat fake news, but instead is likely to backfire and have unintended negative effects by decreasing the belief of and engagement with true news articles.

KEYWORDS

Fake news; online disinformation; news trustworthiness; AI recommendations; deception detection; AI vs. human

Introduction

Interest in fake news on social media was prompted by the 2016 US presidential election, which was influenced by fake news articles spreading on social media [33, 39]. Fake news has gotten worse with COVID [92], and many users say it is difficult to distinguish between true and fake news [5]. Fake news on social media is now recognized as a major societal problem [72].

Most social media platforms use artificial intelligence (AI) to recommend articles that fit the taste of users [34], but the AI recommendation is seldom explicitly disclosed [106]. Surveys show that the majority of social media users are not aware of the role of AI in the curation of their news [29, 76, 103]. As a result, experts advocate that the use of AI to recommend articles should be clearly disclosed because such disclosure may make users more cautious about news articles on social media [5, 56]. Policymakers are beginning to act, with new laws requiring the disclosure of AI recommendations being considered in

several countries.¹ But does the explicit disclosure of AI recommendations reduce belief in and sharing of fake news articles as policymakers hope?

People have long relied on recommendations from other people to inform their choices [8], and research shows that expert and user ratings have significant effects on users' belief in news [50]. Many social media news articles are explicitly recommended by a friend or an expert [48]. Research in other areas shows users trust AI to make recommendations, especially for media [59]. If social media users transfer their trust in AI from other areas to AI recommendations of news, a policy requiring AI disclosure could backfire, and have the opposite effect by *increasing* user's belief in news articles labeled as recommended by AI. Understanding whether disclosing the role of AI agents in recommending news articles on social media increases or decreases belief in fake news or the sharing of fake news is critical in order for government policymakers and social media platforms to make good decisions about the disclosure of AI recommendations.

We used two experiments to investigate the effects of labeling a news article as recommended by AI compared to human recommenders (i.e., an expert, or a friend). In both studies, the net effect of an AI recommendation was to *reduce belief in true articles* and have no material effect on belief in fake articles. Trust in the recommender played an important role, but did not fully explain the differential effect of AI on true and fake news, suggesting that some other theoretical mechanism is also at work. Belief that the article was true increased sharing intentions, but the presence of an AI recommender reduced the importance of this belief, making confirmation bias (the alignment of the article with the users' *a priori* beliefs) the primary factor influencing sharing. Thus, the net effect of labeling stories as recommended by AI has no effect on fake news, harms belief in true news, and makes the decision to share articles depend more on whether the article aligns with prior political opinions than considerations about whether the article is true or false. These are not the effects that policymakers hope or researcher theorize. We conclude that policies requiring disclosure regarding AI recommendations are likely to backfire and *exacerbate* the problem of fake news, and theories about AI recommendation of social media news need to be revised.

Prior Theory and Research

Fake News: Detection, Infection, and Mitigations

The seminal definition of fake news is “news articles that are intentionally and verifiably false and could mislead readers” [2]. This excludes articles that cannot be verified, and articles that are partially true but biased. Likewise, this definition only includes false statements that were created intentionally (not those created unintentionally by accident) regardless of whether the author knows the statements are false or not [56].

There are three main streams of fake news research closely related to this study: detecting fake news, understanding why users believe fake news, and combating fake news [333]. First, scholars have devoted much research to detecting fake news, because detection is key to platforms removing fake news or warning users. Research has examined using expert fact-checkers [73, 78, 112], users [3, 112], and computational methods [109]. Computational methods are fast, cost less and scale better than humans, but evolving and nonstationary fake news features make it challenging. Researchers struggle to balance

detection accuracy and speed [109]. Second, people believe and spread fake news because it fits their existing beliefs [73]. Social media users often use lazy reasoning [80] and are inattentive to accuracy [72, 79]; as a result, news presentation format [49] and modality [98], content sentiment [44], others' comments [17], personal anxiety [54], and digital literacy inadequacy [41] increase the belief in fake news. Third, many interventions to reduce the spread of fake news have been examined [50, 71], such as fact checking [3], interface redesigns [72], fake news flagging [50], social norms injection [52, 74, 107], and literacy training [41]. Fake news warnings are most common but their effectiveness is unclear [71, 78]. Labeling AI curation has been proposed as a way of alerting users [58], but there is no evidence of its effectiveness.

News on social media is different from news on traditional media [27], in that there is often a third-party that recommended the article [99]. A recommendation is a form of endorsement, and affects individuals' beliefs and engagement [47]. For example, your beliefs about an article shared by a friend may be influenced by your attitudes toward that friend and your engagement with the article (e.g., "liking") may be influenced more by your relationship with your friend than the article.

News articles are commonly recommended by embedded AI [34], although few users understand this [29, 76, 103]. Policymakers have come to believe that requiring social media platforms to disclose that articles are recommended by AI will help combat the belief and spread of fake news, but this remains an open question.

Social Influence Theory

Social influence theory argues that individuals' beliefs and behaviors are affected by the influence of others around them [13, 23]. Social influence theory has been used to understand information system use behavior [108], and product quality perceptions [53]. Deutsch and Gerard [23] described two types of social influence: informational influence and normative influence. Informational social influence is persuasion to accept the opinions of others based on the information or evidence they provide [23]. Normative social influence is "the influence to conform to some expectations of other individuals through reward-punishment relationships or a desire on the part of the person to identify with other individuals or their point of view." ([81], p.615). Normative influence affects beliefs by the desire to fulfill the expectations of significant others, while informational influence affects beliefs by the desire to have accurate beliefs [14]. Normative influence may cause people to do something they dislike or agree with something they disbelieve, just to satisfy others and avoid conflict [81].

Informational influence and normative influence are not mutually exclusive; both can operate simultaneously [14]. Individuals' responses vary by the type of influence [83]. For informational influence, the sources that are the most trustworthy have the most influence, because it is the information that influences decisions [67]. For normative influence, the most influential sources are those that are socially important, or who can have the power to reward or punish [53]. Both informational influence and normative influence may affect users' news beliefs [63].

Trust and Social Influence

People frequently receive recommendations in their daily lives, such as those about music, movies, products and so on. On e-commerce sites, AI algorithms give users recommendations on products they may like based on their purchase histories [61]. It is similar for news on social media where stories recommended by friends intermix with stories recommended by AI. We follow prior research [73] and use Facebook as our research context since Facebook is the most popular social media platform. The new stories on Facebook may be true or fake, designed to deliberately influence beliefs and behaviors, and may come from foreign agents or social media friends. When these individuals recommend articles on social media, they are conduits between original publishers and consumers, and filter news. When the article source is unfamiliar, users often rely on the recommender when making judgments [23, 48].

The trustworthiness of the recommender plays a major role [13, 81]. Trust beliefs are influenced by two major factors. The first is people's disposition to trust, which is "akin to a personality trait" ([86], p.651); some individuals are simply more or less likely to trust others [66]. The second factor is an individual's perceptions of a recommender, typically driven by their ability, benevolence, and integrity [65, 66]. Ability, benevolence, and integrity are often associated with humans, but they also apply to AI [22]. The relative importance of ability, benevolence and integrity depends upon the situation. For example, for commercial transactions, ability is usually the dominant factor, because the focus is on the seller's ability to complete the transaction [88].

The relative importance of ability, benevolence and integrity may also differ depending on the nature of the influence process. Informational social influence derives from the human need for accurate beliefs about reality [83], so influence relies on objective reasons (i. e., arguments, evidence). Informational influence is stronger if the source is an expert [1]. Thus, the ability of the source is most important to informational social influence [13, 42] with integrity and benevolence playing lesser roles [6]. As for normative social influence, people usually conform in order to fulfill the real or imagined expectations of significant others, such as friends [14]. Thus, the strength of normative social information is primarily influenced by the identity and importance of the recommender [103], due to individuals' need for social approval. Significant others usually have high levels of emotional trust, which is primarily based on benevolence and integrity [88] since they are associated with emotions and personal feelings (e.g., care, love). Ability may also play a role, but it is usually less important than benevolence and integrity [110].

Research Model

In this paper, we theorize that the person or technology that recommends a news article will influence the extent to which social media users believe the article, which in turn affects users' public engagement with news by liking, commenting and sharing. Based on prior research [42] and current social media technology, we examine three primary sources that may recommend news and thus may exert influence on the receivers' belief in the news, namely an AI agent, an expert, or a friend. Although an AI agent is a nonhuman recommender, people often interact with AI agents similarly to how they interact with humans [42], and thus the ability, benevolence and integrity of AI play an important role in the

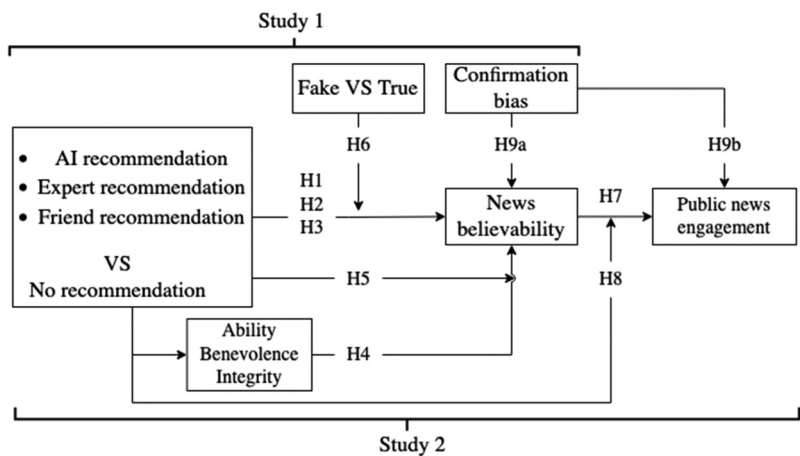


Figure 1. Research model

acceptance of their work [22]. We theorize that the perceived ability, benevolence and integrity of the recommender influence the extent to which users believe recommended articles and the relative importance of the three types of trust on believability will vary by who is the recommender. Our research model is shown in Figure 1.

Artificial Intelligence as a Recommender

An AI recommender refers to any type of autonomous system that uses algorithms to make recommendations [111]. AI is commonly used to make recommendations based on past histories and user profiles in many areas such as music, products, and services [37, 51]. AI is also widely used by social media platforms to curate articles to place in user's news feeds [34].

AI can take many different forms, such as an embedded algorithm or a virtual agent [34]. The perceptions of AI may be different by the representation format, because users are more likely to perceive more human-like virtual agents as social actors than less humanlike embedded AI [20]. A recent review concluded that users trust AI virtual agents more than embedded AI [34]. For example, the virtual representation of a recommendation agent significantly increased users' trust [82]. We theorize and test both forms of AI recommenders in two experiments (a humanlike AI in Study 1 and embedded AI in Study 2). Given past empirical results, the theoretical effects may be stronger for humanlike AI than agentic AI (i.e., stronger in Study 1 than in Study 2) [91, 93, 95].

Our theorizing is about social media users' trust perceptions, so the key focus is not the objective truth, but instead what social media users perceive, because users act on their perceptions not on the underlying facts that gave rise to them [45]. In addition to the differences in ability, benevolence, and integrity triggered by presenting AI in an embedded form versus in a personified form, the context influences users' perceptions of AI [34].

There are two opposite views about the ability of AI. On one hand, AI's superior ability in information processing and pattern recognition has been shown in some areas [97]. On the other hand, people often perceive AI as unknowledgeable [64], and lacking appropriate

Table 1. Comparisons among recommenders on three dimensions of trust

Recommender	Ability	Benevolence	Integrity
AI	Low	Low	Low
Expert	High	Medium	Low/High
Friend	Medium	High	High

Abbreviation: AI, artificial intelligence.

goals [51], and human resistance to AI advice has been documented in several settings [24, 51, 60, 113]. Fake news articles are evolving and becoming more sophisticated [109], so judging the truthfulness of news articles is difficult [100]. When there is an absence of well-defined, comprehensive rules—as is the case with fake news today—it is challenging for AI to generate assessments as reliable as professional experts [18, 100]. Policymakers are convinced that the general public believes AI has a poor ability to recommend news articles, which is why they strongly advocate that articles recommended by AI should be labeled as such [29]. Therefore, we theorize that social media users will perceive AI to have a low level of ability (see Table 1). Again, we note that what matters is not the actual ability of AI, but rather what users believe about AI's ability.

Benevolence is a human characteristic [32]. An AI agent does not have human emotions or empathy, and has no capacity to feel the interpersonal care and concern that is central to benevolence [64]. Benevolence is sometimes tied to expectations of reciprocity, or reciprocal altruism, both of which are also lacking in AI [30, 88]. Therefore, we theorize that users will perceive an AI recommender to have a low benevolence (see Table 1).

Integrity is consistency in adhering to acceptable moral standards [66]. AI itself does not have moral standards in the same way that humans have moral standards [24]. Users often perceive that AI makes decisions by maximizing a predefined outcome of interest to its creator, rather than considering moral issues, so they resist moral decisions made by AI [24]. Users often believe that the AI serves the company's welfare first and their customers' second if at all [34]. Facebook has been accused of bias in news feeds [10, 84], and many Americans have shown distrust toward the platform about political news [36]. The fundamental moral fairness of AI-enabled technologies has been challenged as AI sometimes behaves in a manner that is morally dubious [102]. It is hard for users to believe an AI recommender has high integrity [15]. Therefore, we theorize that users' perception of AI's integrity will be low (see Table 1).

In summary, research on information influence and normative influence shows that the trustworthiness of a recommender plays a major role in accepting recommendations [13, 81]. The trustworthiness of a person (or AI) is strongly influenced by their perceived ability, benevolence and integrity [22, 65, 66], although the relative importance of ability, benevolence and integrity may differ depending on the situation. AI is more likely to exert informational influence than normative social influence, because AI is not a "significant other," so the ability of the AI is likely to be more important than its benevolence and integrity. Given the perceptions of low ability (and low benevolence and integrity), users are likely to have low trust in AI recommendations. Thus, we theorize that news articles labeled as recommended by AI are less likely to be believed.

Hypothesis 1 (H1): Social media users are less likely to believe an article when it is recommended by AI, compared to having no recommender.

Experts as a Recommender

An expert is a highly capable person with special knowledge, skill, or training in a certain domain [70, 96]. Experts are commonly believed to possess high expertise and professional knowledge in their domain of expertise [70, 96]. In this context, we define an expert as an expert at recommending news. Thus, experts are perceived to have high ability in the domain of recommending news; it is this ability and expertise that makes them an expert [70]. See Table 1.

The integrity of experts is more nuanced. Most experts are presumed to adhere to acceptable principles when using their professional expertise [85]. Thus, they would be perceived to have high integrity. However, there is evidence that some social media users believe that professional experts exhibit political bias [31]. Thus, they have very different perceptions of the integrity of news experts depending on the expert's political alignment; experts with the same political alignment as the user are perceived to have high integrity, while experts of the other political alignment are believed to have low integrity (see Table 1). Note that this is distinct from ability because many people who believe these politically biased experts lack integrity do not impugn their ability.

Most people will ascribe some level of benevolence to other humans they do not know, at least more than to machines [30, 88]. However, it is unlikely that users will think that an unknown person sincerely cares about them to the same extent as friends, so benevolence is likely to be higher than AI, but lower than close friends (See Table 1).

As we previously noted, the relative importance of ability, benevolence and integrity may differ depending upon the situation. An expert is more likely to exert informational influence than normative influence, because an expert is not a “significant other.” Some experts may be celebrities (e.g., Dr. Anthony Fauci is a medical expert and a celebrity) and some celebrities may be perceived to be experts when they are not (e.g., Sean Hannity is a celebrity but not a medical expert). We exclude celebrities because theory about celebrities is different from theory about experts. Celebrities have para-social influence [46] and are more likely to have normative influence because they are an “important other,” whereas non-celebrity experts have ability-based influence as they are unlikely to be an “important other” [13, 23].

Thus, for articles recommended by an expert, the ability of the expert is likely to be more important than its benevolence and integrity, although these may play a role. Given the perceptions of high ability, we theorize that users are likely to have higher trust in expert recommenders. Therefore, we theorize that experts' recommendations will have a positive influence on users' belief in the news articles. Thus:

Hypothesis 2 (H2): Social media users are more likely to believe an article when it is recommended by an expert, compared to having no recommender.

Friends as a Recommender

Social media users have many friends, some of whom are close family members and others who are less well known to the user [12]. Most social media friends are neither close nor unknowns, but instead are somewhere between the two ends of the spectrum [12, 77]. Therefore, in this study, we focused on friends who were not close family members with whom the user has a relationship.

Users' social media friends who share news articles are likely to make similar accuracy evaluations as other users—neither more nor less accurate [83]. We exclude friends who are experts. Most social media users find it difficult to distinguish between true and fake news [5, 100], and believe other users are worse than themselves [19]. Thus, users are likely to perceive that their friends have less ability than experts but more ability than AI (Table 1).

Users usually believe their friends are benevolent [83]. Few people maintain relationships with individuals who display a lack of benevolence [68]. Our focus is on the user's perceptions of his or her friends, not the actual benevolence those friends feel toward the user. Most users will perceive that their friends have a higher level of benevolence than strangers they do not know.

Likewise, social media users usually ascribe some moderate level of honesty and integrity to their friends [83]. Few people maintain a relationship with individuals who lack integrity [12, 68]. Thus, most users will perceive that their friends have a higher level of integrity than strangers they do not know. Users may have different benevolence and integrity perceptions for different friends.

A recommendation from a friend may carry both informational influence and normative influence because users are more likely to have attachments and commitments to their friends [53] and friends are more likely to be an "important other" [13, 23]. Thus, the friend's benevolence and integrity may be more influential than the friend's ability. Normative social influence may also exert influence to believe the friend's recommendation in order to maintain the social relationship with the friend [14, 53, 81]. Therefore, recommendations from friends will have a positive effect:

Hypothesis 3 (H3): Social media users are more likely to believe an article when it is recommended by a friend, compared to having no recommender.

Trust as an Underlying Theoretical Mechanism

The theory underlying our hypotheses argues that there is a moderated-mediated relationship. First, individuals perceive that the different recommenders have different ability, benevolence and integrity, and thus these three factors mediate the relationship between recommender and belief. Second, the relative importance of ability, benevolence and integrity depends on the recommender because human experts and AI exert information influence, making ability more important, while friends exert normative influence making benevolence and integrity more important. Thus, we hypothesize that:

Hypothesis 4a (H4a), Ability, 4b (H4b) benevolence, and 4c (H4c) integrity perceptions of the recommender mediate the relationship between news recommendation and news believability.

Hypothesis 5 (H5): The effect of ability will be stronger for a) AI and b) expert recommenders, and the effect of c) benevolence and d) integrity will be stronger for friend recommender.

News Veracity

News veracity is the actual truthfulness of a news article. The news article is either verifiably true or verifiably false; articles that cannot be verified as true or false lie beyond the scope of this study. Social media users often struggle to figure out whether

to believe an article or not—that is, whether to conclude an article is true or false [5]. The actual veracity of a news article is a key factor influencing whether users believe it or not [57]. True articles are more likely to have stronger information influence than fake articles (because they are actually true), and thus true articles are more likely to be believed than fake articles [72, 79]. Nonetheless, some users believe that fake articles are true, and many have difficulty discerning true news from fake news [69]. This is because social media users tend to consider information they encounter as true unless they have reasons to doubt them and invest additional deliberation [49]. Social media users are often in hedonic mindsets, aiming to have fun [7], so they fail to think about whether an article is true or false [80]. If users engage in sufficient analytic reasoning, they are more likely to identify false news stories [71, 80], but social media users prefer to invest minimal effort [71, 80].

A recommendation from a third party provides additional evidence that the article is true over and above the content of the article [104]. A third party recommendation makes users more likely to rely on the recommendation and presume the article is true [9]. A recommendation reduces users' motivation to invest their own cognitive effort [9]—effort that is critical to identifying fake stories [71, 80]. As theorized in the previous section, a recommendation from an expert or a friend will have a positive effect on believability, which is likely to have a differential effect for true versus fake articles. For an article that is true, the effects of relying on the positive recommendation from an expert or a friend instead of investing effort into one's own assessment are likely to have little effect, because both are likely to produce a similar belief that the article is true. However, for an article that is false, the effects of relying on this positive recommendation (which suggests the article is true) instead of investing effort into one's own assessment (i.e., more likely to result in a conclusion that the article is false or uncertainty about its veracity), produce different results. Therefore, the effect of a positive recommendation from an expert or a friend has a stronger effect on believability for verifiably false articles than for verifiably true articles.

In contrast, as we previously argued, a recommendation from AI reduces believability. Since fake news articles will have lower believability, both reliance on AI recommendations and users' deliberation will produce similar negative effects on believability. However, for a true news article, users who invest cognitive effort will have higher believability, while relying on the AI recommendation would lead to lower believability. Thus, the negative effect of AI recommendation will be stronger for true news articles. Hence:

Hypothesis 6 (H6): The effect of a) AI recommendation on believability will be stronger for true news articles than fake news articles, while the effect of b) expert, and c) friend recommendation on believability will be stronger for fake news articles than true news articles.

News Engagement

News stories with high believability are more likely to be shared [73], although people do share news articles they know to be fake [101]. This is because people prefer to share news stories that they believe are true and tend to avoid liking or sharing something that appears fake and ridiculous [73]. Believability is an antecedent of users' intention to engage with social media news stories because they are more likely to like and share stories they believe to be true [71]. Thus,

Hypothesis 7 (H7): Social media users are more likely to engage with an article they believe is true.

News recommendation is a publicly visible external cue attached to the article which can change the weight users put on belief about the article itself when they decide whether to take action or not. If the user believes the recommender has strong normative influence or strong informational influence then the weight given to the user's own belief in the article may become less important when deciding whether to publicly engage with the article.

When faced with an article recommended by a close friend who exerts strong normative influence, the decision whether or not to like or share the article may depend on the relationship with the friend rather than the user's beliefs about the veracity of the article [35]. Friends are likely to exert normative social influence, which reduces users' attention on the information content and shifts users' attention to factors such as relationships and self-identity [27]. When users receive a news recommendation from a good friend, the impacts of other factors (e.g., interpersonal relationship maintenance, impression management) would play a more important role [26]. Thus, the relationship between news believability and intentions to act will be weakened by a friend recommendation, because sharing is influenced by normative influence.

Conversely, when deciding whether to share an article recommended by an expert, users will feel less normative influence. They may rely more on their assessment of the expert's ability. As theorized in the prior sections, and as found empirically in Study 1, users perceive experts to have high ability to recommend news and are susceptible mostly to their informational influence. When human experts recommend news, users tend to focus more on information, which means users' belief in the information's credibility would have a larger effect on the decision to share than the relationship with the expert. Therefore, the relationship between believability and intentions to act will be strengthened by expert recommendations.

A recommendation by AI could be perceived as a third-party endorsement of an article by the social media platform providing the AI similar to a platform-sanctioned fact-checker opinion [38, 62]. Such endorsements may also be seen as a substitute for one's own belief in the article. In this case, AI recommendation will weaken the connection between the user's belief and the user's engagement with a news article. Therefore, we hypothesize:

Hypothesis 8 (H8): The effect of news believability on users' engagement with a news article will be weaker when it is recommended a) by AI or c) by a friend, and stronger when it is recommended b) by an expert, compared to when it is not recommended.

Confirmation Bias

One other important factor explaining fake news spreading is confirmation bias, which means people are more likely to believe information aligned with their own point of view than information that opposes it [73]. Human decision-makers prefer information matching their preexisting opinions, and dislike information challenging them [75], likely because accepting information that aligns with preexisting opinions is cognitively easy and emotionally comfortable [114]. People unconsciously accept information consistent with their prior beliefs as true, without being aware of this bias [49]. In contrast, encountering a piece of challenging information can cause cognitive dissonance [49] which requires additional

cognitive effort to resolve, so people tend to reject this opposing information and retain their opinions [73]. Past research suggests that confirmation bias is one reason why social media users believe fake news even when they are flagged as fake [499, 73]. Confirmation bias makes users more likely to believe social media news articles supporting their political party [49, 73]. Thus, we hypothesize:

Hypothesis 9a (H9a): Social media users are more likely to believe an article aligned with their opinions.

Confirmation bias can also influence social media engagement [49]. News believability and confirmation bias are the primary drivers of user behaviors [49] as users tend to like, comment and share news stories supporting their preexisting beliefs. Although confirmation bias can influence users' assessment of news believability, confirmation bias has impacts on user behavior over and above the effects of news believability because users share news to present self-identity and follow the social norms of a certain community or group [55]. Thus, we hypothesize that users are likely to like, comment on or share news articles aligned with their views of the external world.

Hypothesis 9b (H9b): Social media users are more likely to engage with an article aligned with their opinions.

Study 1: News Belief with Humanlike AI Agents

Participants

We recruited 235 American participants from the Cloud Research Mechanical Turk panel; Cloud Research verifies MTurk participants to ensure they are real people, live in the country they claim, and provide quality data. Online survey platforms like Cloud Research, MTurk, and Prolific Academic are widely used in fake news studies [71, 78] because they provide participant panels similar to the general population that use social media and have at least as good quality data as student samples [11]. About 60 percent of the participants were male, and 40 percent of our participants were female. Their age ranged from 18 to more than 65: 3 percent were less than 24, 38 percent between 25 and 34, 34 percent between 35 and 44, 12 percent between 45 and 54, 10 percent between 55 and 64, and 3 percent were older than 65. The participants were diverse in education: 14 percent of the participants didn't attend college, 14 percent attended college but didn't have a degree, 7 percent had a two-year degree, 54 percent finished a four-year degree program, and nearly 11 percent completed a graduate degree; this sample is slightly more educated than the U.S. population [87]. More than 68 percent were daily Facebook users (19 percent used it once a week). About 54 percent were Democrats, 37 percent Republicans, and 9 percent independents.

Task

This experiment took about 15 minutes. Participants read eight headlines and answered questions measuring their belief. The headlines were assessed in two pilot studies which showed that the materials had no major differences in the type and magnitude of feelings (see Online Supplemental Appendix 1 and 1.3). Half of the headlines were verified as true

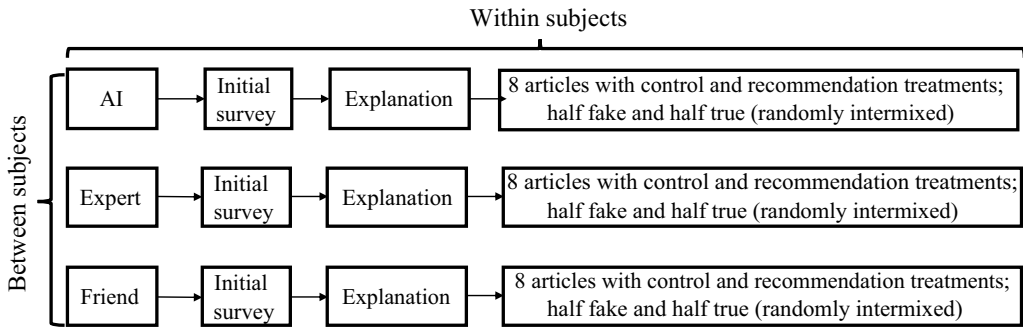


Figure 2. Design of the experiment

news with high perceived plausibility, and half were fake news; half appealed to Democrats, and half to Republicans.

Treatments

The experiment used a within-subject and between-subject research design (see Figure 2). The type of recommender was the between-subjects treatment, which means that each participant was randomly assigned to receive recommendations from only one type of recommender (AI, expert, or friend). We compared the characteristics of participants in the three treatments (i.e., age, education, and Facebook usage) and found no significant differences, so we conclude that the randomization was successful. A within-subject design is effective in controlling individual-level differences [40]. The within-subject portion was that all subjects saw four headlines in a control treatment with no recommender, and four headlines in their assigned recommender treatment. In addition, the headlines are counter-balanced in accuracy, with four headlines true and four headlines false. Headlines were randomly assigned to the control or recommender treatment and presented in random order, with the two intermixed.

The headlines in the control condition were displayed as they would appear on Facebook, with an image, a headline, and a media outlet; the media outlet was different for each headline. In the recommender treatments, a recommender was attached to the headlines, with other elements identical to the control treatment. As a manipulation check, the participants were asked to recall who recommended news to them after assessing the eight headlines. We removed the data of those who did not correctly recall the recommender. All headlines with a recommender had both an icon and text about the recommender, as is usual on Facebook (see Online Supplemental Appendix 2.2).

To control for any gender, race/ethnicity, age, and/or appearance effects, the name and image of the recommender were held constant for all headlines. This decision to provide experimental control, following the established practice in prior studies [25], led to a weaker friend treatment, because the friend recommendations did not include the actual names and images of the participant's friends. We believed that asking for names and images of friends could cause privacy concerns and reduce experimental control due to potential differences in ability, benevolence and integrity for different friends. We used a photograph for both

human treatments (expert and friend) and a matching caricature for the AI treatment because AI cannot have a photograph and avatars are well accepted representatives of technology-based agents. Thus, we kept the AI treatment similar to the friend or expert treatment; if the AI treatment did not have a similar image, then it would introduce a potential confound because we would not know if the differences were due to AI or the lack of an image. A separate page with a welcome image was presented to explain the recommender before participants saw the headlines, as shown in the Online Supplemental Appendix 2.1. We conducted a pilot test to ensure that the explanations of the three recommenders were suitable and well understood; minor changes were made to the explanations.

Measurements

News veracity was defined by the materials. Headlines were verified as true or verified as false. We measured confirmation bias as the alignment of the headline with the participant's political beliefs following the process of Pennycook et al., [78]. Headlines were selected, pretested, and confirmed as being consistently pro-Democrat (-1) or consistently pro-Republican (+1); headlines that were not consistent in political leaning were not used (see details in the Online Supplemental Appendix 1). Following the process of Pennycook et al. [78], we multiplied the headline's leaning (-1 or +1) by the participant's party identification (-3 to 3; strong Democrats to strong Republicans) to capture the headline's alignment with the participant's political identification (which was measured following Moravec et al. [71 72]). We used three items adapted from Moravec et al. [73] to measure news believability. We standardized all the variables for analysis. See the Online Supplemental Appendix 3 for the items.

Results

We used a multilevel mixed-effect linear regression with random intercepts in Stata to analyze the effects of the treatments on news believability following the approach of Baron and Kenny [4]; we could not use the Hayes Process [43] because it does not support multiple observations per participant or the MLMED macro because it does not support multiple treatments. Multilevel mixed-effect linear regression is used when there are multiple observations per subject (e.g., [71]). The control group was the baseline with no recommendation.

The first model is the direct model (Step 1 in Table 2) that shows the overall effects (see Online Supplemental Appendix 4 for the full statistical tables). The explanatory power of this model ($R^2 = .22$) is what Cohen [16] calls a "medium" effect size (i.e., "visible to the naked eye" [16:26]). The results show that AI recommendation had a negative and significant effect on news believability, but there is also a significant interaction between AI recommendation and the veracity of the article. Expert recommendation had a significant positive effect. The effect of friend recommendation was not significant. Thus, H2 is supported, H3 is not supported, and H1 is supported in some conditions, but not others (see the results of H6a in the following section).

H6a-c argued that the effect of recommender on believability would be different for fake news articles, compared to true news. The interaction between fake articles and

Table 2. Impacts of recommendation on news believability and mediators in Study 1

Dependent variable:	Step 1: Direct model	Step 2: Mediators			Step 3: Mediated model
	Believability	Ability	Benevolence	Integrity	Believability
AI	-0.196*	0.111	-0.203	-0.024	-0.189*
Expert	0.164*	0.347*	-0.155	0.052	0.138
Friend	-0.024				-0.027
Fake Article	-0.835***				-0.832***
AI X Fake Article	0.238*				0.236*
Expert X Fake Article	-0.130				-0.134
Friend X Fake Article	0.001				-0.002
Confirmation Bias	0.318***				0.316***
Ability					0.112*
Benevolence					0.180**
Integrity					0.068
AI X Ability					-0.065
AI X Benevolence					-0.021
AI X Integrity					0.005
Expert X Ability					0.007
Expert X Benevolence					-0.093
Expert X Integrity					0.033
Control variables					
Female	-0.123	-0.017	-0.111	0.007	-0.108
Education	0.039	-0.010	0.070	0.075	0.027
Age	0.033	0.038	0.067	0.089	0.007
Facebook	-0.136***	-0.369***	-0.391***	-0.348***	-0.008
Constant	0.466***	-0.169	0.161	-0.021	0.465***
Observations	1880	234	234	234	1872
R2 (Percent)	22.2	14.2	18.9	14.4	38.7

Note: *p < 0.05. **p < 0.01. ***p < 0.001. Abbreviation: AI, artificial intelligence.

recommender was only significant when the recommender was AI. The main effect of a recommendation from AI on believability was negative and significant, while the interaction between AI recommendation and fake articles was positive and significant, which means the negative effect of AI recommendation was weaker for fake news. The interactions between fake news articles and human recommendations were not significant; H6a was supported, but H6b and H6c were not.

The beta coefficients for the main effect of AI recommender (-.196) and its interaction with fake news articles (.238) suggest that these are equal offsetting effects, which means that AI recommender only affects true news. As a robustness check, we conducted subsample analyses to understand the effects of each recommender on fake versus true news separately. This involves two separate analyses [28], one for the true articles and one for the fake articles, which can more clearly show the effects for the two different types of news articles compared to using interaction terms [90]. This analysis (See Table 3) shows that the effect of AI recommenders on beliefs is negative and significant for true news ($b = -.150$, $p = .033$) and has no significant effect for fake news ($b = -.006$, $p = .940$). Although the interaction between expert recommendation and fake articles was not significant in the main analysis (see Table 2), the subsample analysis shows a differential effect for expert recommendation: there is a significant positive effect of expert recommendation on news believability for true news but none for fake news (see Table 3). Thus, combining the results of the main model (Table 2) with the subsample analysis (Table 3), we conclude that the positive effect of expert

Table 3. Subsample analysis on news believability (Study 1)

	Coef.	Std. Err.	T	p	95% Confidence interval	
Table 3a: True news articles						
AI	−0.150*	0.070	−2.140	0.033	−0.287	−0.012
Expert	0.133*	0.067	1.990	0.046	0.002	0.264
Friend	−0.034	0.063	−0.530	0.593	−0.158	0.090
Confirmation Bias	0.188***	0.044	4.290	0.000	0.102	0.274
Female	−0.058	0.079	−0.730	0.463	−0.212	0.096
Education	0.038	0.036	1.040	0.297	−0.033	0.109
Age	0.068	0.037	1.860	0.063	−0.004	0.140
Facebook	−0.069	0.039	−1.780	0.075	−0.146	0.007
Constant	0.436***	0.053	8.220	0.000	0.332	0.540
Table 3b: Fake news articles						
AI	−0.006	0.082	−0.070	0.940	−0.168	0.155
Expert	0.067	0.078	0.850	0.393	−0.087	0.220
Friend	−0.015	0.075	−0.200	0.845	−0.161	0.132
Confirmation Bias	−0.001*	0.044	−0.020	0.980	−0.087	0.085
Female	−0.204	0.046	−4.380	0.000	−0.295	−0.113
Education	−0.339	0.063	−5.370	0.000	−0.462	−0.215
Age	−0.006	0.082	−0.070	0.940	−0.168	0.155
Facebook	0.067	0.078	0.850	0.393	−0.087	0.220
Constant	−0.015	0.075	−0.200	0.845	−0.161	0.132

Note: * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. Abbreviation: AI, artificial intelligence

recommendation on believability is supported for true news articles, but its effects for fake news articles are not consistently supported.

As a post-hoc test, we examined the effects of the recommenders without considering the differential effects of true and fake articles. The results (See Online Supplemental Appendix 4, Table 4.1) show that omitting news article veracity hides the effects of AI recommendations (because the effects of the recommenders are not significant). Thus, failing to consider the differential effects of news veracity obscures the effects of the recommenders.

In Table 2 (direct model), confirmation bias is significant and positive; as a result, H9a was supported.

Theoretical Mechanism Analyses

The test for moderated mediation adds two additional steps to the baseline model in Table 2 [4, 43]. The second step is to test whether there is a significant relationship between the independent variable (recommender) and the mediators (ability, benevolence, and integrity); that is, do users have different perceptions of the ability, benevolence, and integrity of different recommenders? We conducted three separate regression analyses (using friend as the base case) and found that experts were perceived to have greater ability than friends, and AI was perceived to have no significant difference with a friend for all three (see Step 2, Table 2).

The third step is to assess whether the independent variable (recommender) has significant effects on the dependent variable (believability) in the presence of the mediators (ability, benevolence, and integrity) and to assess whether the independent variable (recommender) moderates the effects of the three mediators on the dependent variable (believability). Ability, benevolence and integrity were standardized for this analysis (mean = 0), and their values were set to zero for the control treatment with no recommendation. To keep the baseline group consistent with the analysis in step 2,

the trust perceptions of friend were used as the base case for ability, benevolence, and integrity.

This analysis (see Step 3, Table 2) showed that AI has a significant effect but expert recommender is no longer significant. Ability and benevolence both had significant effects on belief with no differences in relative importance due to different recommenders. The explanatory power of this model ($R^2 = .38$) is what Cohen [16] calls a “large” effect size). This explanatory power is also greater than what has been typically found in past research examining belief of articles, which usually ranges from 10 percent to 20 percent (e.g., [21, 72, 78, 79]). The large effect size of the model suggests that the influence of the recommender is meaningful, as well as significant.

Thus, we conclude that the underlying mechanisms of news recommenders on news belief are complex in that there is a full or partial mediation by perceptions of the recommender’s ability, depending upon the recommender. The effects of expert recommendations are fully mediated by perceptions of the expert’s ability. In contrast, while AI’s ability and benevolence also influence the effects of AI recommendations on belief, there are other factors about AI that influence the effects of AI recommendations on belief. Moreover, the interactions between the recommender and trust perceptions were all not significant. Thus, H4 was partially supported, but H5 was not supported. The negative effect of labeling social media stories as recommended by AI is not fully explained by the three trustworthiness mediators.

Discussion

The results in Study 1 show that labeling a social media story as recommended by AI has the net effect of *reducing* belief in true news, but has no material effects on fake news. Thus, labeling a news article as recommended by AI backfires and harms true stories without affecting fake ones. Ability played a role in belief, but does not explain the asymmetrical effects of AI recommendations. In contrast, labeling a news article as recommended by an expert *increased* believability for true news articles (and possibly for fake news articles), and this effect is fully mediated by the greater perceived ability of the expert. Interestingly, we found no significant effects for the friend recommendations.

Study 2: News Belief and Engagement with Embedded AI

Study 2 extends Study 1 by investigating users’ intention to engage with the news articles (i.e., comment, like, and share). Study 2 also used a different form of AI. Study 1 used a humanlike design to present the AI as a virtual agent. Studies have shown that making AI agents more humanlike may change the impact of AI [34]; therefore, we wanted to examine the effects of AI recommendation for embedded AI that is less humanlike.

We also took this opportunity to change our experimental design in terms of how the friend treatment was presented to participants and how we measured confirmation bias. Study 1 used a generic presentation of a social media friend which might underestimate the impact of friend recommendation; as a result, in Study 2, we asked participants to identify a *specific* social media friend who was used in the experiment. Study 1 followed prior studies [78] by measuring confirmation bias as the alignment between the political stance of the participant and the political stance of the news article, whereas Study 2, measured

confirmation bias by directly asking users about the alignment between the article and their own point of view, which is also commonly used in past research [50].

Method

The method for Study 2 was identical to that of Study 1, with five changes (two theoretical, three methodological). First, we changed the presentation of AI to an AI algorithm, without any humanlike cues (see the Online Supplemental Appendix 2). This enables us to see the extent to which the results from Study 1 hold for embedded AI as opposed to more humanlike AI. Second, we added three questions to measure news engagement drawn from prior research [71]: I'd like to "Like" it/I'd like to comment on it/I'd like to share it. The Cronbach's alpha was 0.91.

We made three methodological changes. First, we changed the friend treatment to make it more relevant. At the beginning of the experiment, we informed participants that news recommendations from friends are common on social media and we asked them to think of a social media friend and enter his or her first name. This name was then used as the friend treatment. Second, confirmation bias was measured by one 7-point question presented with each news article: "This story matches my opinions on the topic." Finally, we used new news articles to ensure they were relevant, all from the same media outlet we invented (Todaynews.com). We tested 16 news headlines from Snopes.com (half true and half false) that were common at the time of the study and selected 8 headlines (see Online Supplemental Appendices 1 and 2). We also conducted a separate pilot test examining familiarity with AI, and found that participants with greater AI familiarity perceived AI recommenders to have significantly greater ability ($p < .001$), benevolence ($p < .001$), and integrity ($p < .001$); see Online Supplemental Appendix 5.

We recruited 332 participants in the United States from the Cloud Connect platform. Cloud Connect is a platform that provides a reliable subject pool for social science research that is widely used by scholars [89]. About 42 percent of the participants were female, and 58 percent of our participants were male. The age of our participants ranged from 18 to more than 65: 5 percent were less than 24-years-old, 38 percent were between 25 and 34, 31 percent between 35 and 44, 15 percent between 45 and 54, 8 percent between 55 and 64, and 3 percent were older than 65-years-old. 11 percent of the participants did not attend college, 20 percent attended college but didn't have a degree, 9 percent had a two-year degree, 43 percent finished a four-year degree program, and nearly 16 percent completed a graduate degree; this sample is slightly more educated than the U.S. population [87]. More than 65 percent were daily Facebook users.

Results

We retested H1-6 and H9a with the new embedded AI and revised methods (see Step 1 in Table 4; see Online Supplemental Appendix 4 for the full statistical tables). AI recommendation again has a negative main effect combined with a significant positive interaction. Expert has a significant positive main effect combined with a significant negative interaction effect, meaning its effects depend on the veracity of the news article. Friend recommendation has a significant positive effect. H3 is supported, but H1 and H2 are supported only in some conditions. H3 was not supported in Study 1

Table 4. Impacts of recommendation on news believability and engagement in Study 2

Dependent variable	Believability				Engagement	
	Step 1: Direct model	Step 2: Mediators			Step 3: Mediated model	Step 4: Direct model
	Believability	Ability	Benevolence	Integrity	Believability	Engagement
AI	-0.249**	-0.592**	-1.680***	-0.763***	-0.282**	0.051
Expert	0.198*	-0.412*	-1.105***	-0.706***	0.204*	0.053
Friend	0.196*				0.190*	0.020
Fake Article	-0.546***				-0.544***	0.050
AI X Fake Article	0.264*				0.264*	-0.053
Expert X Fake Article	-0.262*				-0.263*	-0.013
Friend X Fake Article	-0.101				-0.102	0.051
Confirmation Bias	0.999***				0.994***	0.371***
Ability					0.120*	
Benevolence					-0.066	
Integrity					0.060	
AI X Ability					-0.067	
AI X Benevolence					-0.144	
AI X Integrity					0.212*	
Expert X Ability					0.038	
Expert X Benevolence					0.089	
Expert X Integrity					-0.076	
Believability						0.256***
AI X Believability						-0.122*
Expert X Believability						0.112*
Friend X Believability						0.004
Control variables						
Female	0.152*	0.151	0.018	-0.024	0.138*	-0.330**
Education	-0.076*	0.020	0.054	0.029	-0.073*	0.139*
Age	-0.052	-0.053	-0.081	-0.032	-0.047	-0.135*
Facebook	-0.027	0.138	0.192*	0.111	-0.036	0.077
Constant	4.354***	4.885***	4.623***	5.202***	4.356***	2.548***
Observations	2656	332	332	332	2656	2656
R2 (Percent)	47.6	3.7	18.5	8.5	54.8	25.2

Note: *p < 0.05. **p < 0.01. ***p < 0.001. Abbreviation: AI = artificial intelligence.

where we used the generic friend treatment, but when participants thought of a specific friend, it was. This suggests that the more realistic friend treatment may be a stronger test of the hypothesis. The interaction between AI recommendation and fake news was again positive and significant and the interaction between friend recommendation and fake news was not significant (H6a supported and H6c not supported), which is the same as the results in Study 1. However, H6b is not supported as we found a significant *negative* interaction between expert recommendation and fake news, not the hypothesized *positive* interaction; the results are the reverse of what we hypothesized.

As a robustness check, we again conducted two separate subsample analyses for fake news or true news and found that the effect of AI recommenders on belief was again negative and significant for true news articles ($b = -.175$, $p = .004$) and had no effect for fake news articles ($b = -.019$, $p = .730$); see Table 5. The positive effect of expert recommendation on true news believability was found again ($b = -.130$, $p = .031$), while fake news believability was not affected ($b = -.053$, $p = .343$), which is consistent with the results of subsample analyses in Study 1. Moreover, the confidence intervals for the coefficients on AI

Table 5. Subsample analysis on news believability (Study 2)

	Coef.	Std. Err.	T	p	95% Confidence interval	
Table 5a: True news articles						
AI	-0.175**	0.060	-2.910	0.004	-0.293	-0.057
Expert	0.130*	0.060	2.160	0.031	0.012	0.248
Friend	0.139*	0.059	2.350	0.019	0.023	0.256
Confirmation Bias	0.595***	0.022	27.310	0.000	0.552	0.638
Female	-0.019	0.025	-0.780	0.436	-0.068	0.029
Education	0.076	0.051	1.500	0.133	-0.023	0.175
Age	-0.050*	0.025	-2.020	0.043	-0.099	-0.002
Facebook	-0.029	0.025	-1.170	0.243	-0.077	0.020
Constant	0.129**	0.038	3.360	0.001	0.054	0.204
Table 5b: Fake news articles						
AI	0.019	0.056	0.340	0.730	-0.090	0.129
Expert	-0.053	0.056	-0.950	0.343	-0.163	0.057
Friend	0.056	0.055	1.010	0.313	-0.052	0.163
Confirmation Bias	0.668***	0.020	33.510	0.000	0.629	0.707
Female	-0.044	0.025	-1.800	0.072	-0.093	0.004
Education	0.110*	0.050	2.200	0.028	0.012	0.208
Age	-0.043	0.024	-1.760	0.079	-0.091	0.005
Facebook	-0.003	0.024	-0.140	0.887	-0.051	0.044
Constant	-0.227***	0.037	-6.120	0.000	-0.299	-0.154

Note: * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$. Abbreviation: AI, artificial intelligence.

recommendations overlap between Study 1 and Study 2 (see [Tables 2 and 5](#)), which suggests that the humanlike AI design and the agentic AI design have similar effects.

We again did a post-hoc analysis of the effects of the recommenders without considering the differential effects of true and fake articles. The results (see Online Supplemental Appendix 4, Table 4.4) again find that if news article veracity is not considered, the effects of the recommenders are hidden.

[Table 4](#) shows that confirmation bias was significant; therefore, H9a was supported again.

Next, we proceeded to test the moderated-mediation by ability, benevolence and integrity. As shown in Step 2 [Table 4](#), the ability, benevolence, and integrity of both AI and expert recommender were all significantly lower than those of the friend recommender. We conducted Wald tests to examine the differences between AI and expert recommender and found that the only significant difference was that expert was rated as more benevolent than AI ($p = 0.007$).

Step 3 in [Table 4](#) shows that perceptions of the recommender's ability have significant effects on news believability (matching the results from Study 1), but benevolence and integrity were not significant. There was a significant interaction between integrity and AI recommendation, indicating that perceptions of integrity were important for an AI recommender. None of the other interactions were significant indicating that the effects of ability, benevolence, and integrity were not different between expert and friend recommenders. The direct and interaction effects of AI, expert, and friend recommendation were still significant after including the three mediators, indicating that they are partial mediators. H4 was partially supported and H5 was not supported.

Step 4 in [Table 4](#) shows that news believability is a significant predictor of users' news engagement. All three types of news recommendation had no significant effects on engagement after including news belief, indicating that believability fully mediated the relationship

between recommendation and news engagement, which supports H7. The interaction between AI recommendation and news believability was negative and significant, while the interaction between expert recommendation and news believability was positive and significant, which supports H8a and H8b. An AI recommendation weakens the effect that belief that a news story is true has on engagement while an expert recommendation strengthens the effect. Confirmation bias has a direct and positive effect on news engagement; H9b is supported.

Discussion

We initially noticed that most of the main findings in Study 1 are supported in Study 2, even though we changed the experimental design, showing the robustness of our results in Study 1. The results in Study 2 again show that who or what recommends a news article affects users' belief. AI recommendation once again reduced belief in true news, but had no material effects on fake news. Thus, labeling a news article as recommended by AI backfires and harms true stories without reducing belief in fake ones. Once again, ability played a role in belief, but this was not the reason for the asymmetrical effects of AI recommendations. As in Study 1, labeling a true news article as recommended by an expert again increased its believability, but labeling it as recommended by a friend had a positive effect in Study 2. Perceptions of the ability and integrity of the AI partially mediate the effects of AI recommendation of belief.

One important finding is that an AI recommendation made belief that an article was true *less* important in the decision to share. In other words, flagging an article as recommended by AI reduced the importance that the user believed the article was true, making confirmation bias a stronger predictor of engagement; simply put, flagging an article as recommended by AI increases the likelihood that users will share articles that align with their beliefs even when they are not sure if the articles are true. Conversely, flagging an article as recommended by an expert had the opposite effect—it made belief a more important factor compared to confirmation bias.

General Discussion

This study shows that proposed policy initiatives intended to reduce the belief in and spread of fake news by labeling social media stories as recommended by AI are likely to backfire and instead harm true news articles without reducing belief in fake news articles. We summarize the results of hypotheses testing in Table 6. The results show that across both studies, flagging an article as recommended by AI had no effect on the belief of fake articles, but *decreased* belief in true articles. The effects were the same when the AI recommender was an AI agent with humanlike elements in Study 1 or embedded AI with a plain, machinelike display in Study 2. Thus, the asymmetrical influence of AI recommendations generalizes to both forms of AI.

The mechanism analysis shows that the perceived trustworthiness of the AI recommender (i.e., ability, benevolence, and integrity) is a partial mediator; that is, trustworthiness plays a role in the asymmetrical influence of AI recommendations, but because the direct effects of AI persist in the presence of these mediators, trustworthiness is only part of the explanation. Thus, labeling a news story as recommended by AI recommendation influences also belief through a different theoretical route than by the perceived ability,

Table 6. Results of hypothesis testing

Hypothesis	Study 1 Conclusions	Study 2 Conclusions
H1: Social media users are less likely to believe an article when it is recommended by AI, compared to having no recommender.	Supported for true news articles but not supported for fake news articles.	Supported for true news articles but not supported for fake news articles.
H2: Social media users are more likely to believe an article when it is recommended by an expert, compared to having no recommender.	Supported for true news articles and possibly supported for fake news articles.	Supported for true news articles but not supported for fake news articles.
H3: Social media users are more likely to believe an article when it is recommended by a friend, compared to having no recommender.	Not Supported	Supported
H4: a) Ability, b) benevolence and c) integrity perceptions of the recommender mediate the relationship between news recommendation and news believability.	Partial Mediation a) Supported b) Not Supported c) Not Supported	Partial Mediation a) Supported b) Not Supported c) Supported only for AI
H5: The effect of ability will be stronger for a) AI and b) expert recommender, and the effect of c) benevolence and d) integrity will be stronger for friend recommender.	Not Supported	Not Supported
H6: The effect of a) AI recommendation on believability will be stronger for true news articles than fake news articles, while the effect of b) expert, and c) friend recommendation on believability will be stronger for fake news articles than true news articles.	a) Supported b) Not Supported c) Not Supported	a) Supported b) Not Supported (Opposite Results) c) Not Supported
H7: Social media users are more likely to engage with an article they believe is true.		Supported
H8: The effect of news believability on users' engagement with a news article will be weaker when it is recommended a) by AI or c) a friend, and stronger when it is recommended b) by an expert, compared to when it is not recommended.		a) Supported b) Supported c) Not Supported
H9a: Social media users are more likely to believe an article aligned with their opinions.	Supported	Supported
H9b: Social media users are more likely to engage with an article aligned with their opinions.		Supported

Note: Hypotheses H7, H8, and H9b were not tested in Study 1. Abbreviation: AI, artificial intelligence.

benevolence, and integrity of the AI. In short, contrary to our expectations, the trustworthiness of the AI recommender does not fully explain why AI recommendation has an asymmetrical effect on the belief in news articles.

Engagement with fake news was driven by confirmation bias and belief that a news story was true. Labeling a story as recommended by AI reduced the importance of believing that the story was true, making confirmation bias the driving force in deciding to engage with the news story. Thus, AI labels have no positive benefits in reducing the spread of fake news, but may make the problem worse by increasing the strength of confirmation bias relative to belief that a story is true.

Conversely, recommendations from experts had significant positive effects on belief of true news articles (effects for fake news articles were not consistent). This effect was mediated by the perceived ability of experts in Study 1. Thus, we conclude that social media users trust the recommendations of experts for news, and are willing to take their advice because they perceive the experts to have the ability to select true articles. The effect of a friend recommendation was stronger in Study 2 when users considered a specific friend than a generic friend in Study 1. Once again, ability played a role for recommendations from friends.

Implications for Research

First, and foremost, this paper uses social influence theory to show that users' belief in social media news articles is influenced by who recommends it, extending the current understanding of social media users' belief of fake news. AI recommendations can influence the belief and sharing of fake and true news via some of the same theoretical mechanisms as human experts and friends. The trustworthiness of the recommender was one factor influencing our results, primarily the perceived ability to identify fake and true news. Perceptions about the integrity of AI—or more specifically, its lack of integrity compared to friends—played a role in Study 2. Thus, we need more research on the perceptions of AI trustworthiness for news recommendations. We used experiments in this study, so to further validate the findings, future studies should conduct analyses of real-world data once AI recommendation flagging is implemented.

Second, the effects of AI recommendations are *not* fully explained by the perceptions of its trustworthiness (i.e., ability, benevolence, and integrity). Thus, the negative effect of AI on belief in true articles cannot be explained by the usual factors influencing the trustworthiness of a source. We need more research to better understand the other theoretical processes by which the asymmetrical effects of AI recommendation occur. We speculate that in the hedonic world of social media [7, 71, 80], the presence of an AI recommender is a source of social influence that discourages users from investing their own cognitive effort in assessing the truthfulness of the story [9, 73]. As a result, factors that usually strengthen or weaken belief (i.e., the ability, benevolence, and integrity of the recommender) have lesser effects because users do not actively consider them—they simply accept the recommendation at face value. More studies are needed to better understand the theoretical mechanisms by which AI recommendations act.

Third, users' engagement with fake news (e.g., liking, commenting, and sharing) was susceptible to social influence from a recommendation by modifying news believability and through modifying the impacts that news believability had on engagement. Belief that the story was true had about the same effect as confirmation bias in influencing users' engagement. However, the presence of an AI recommendation reduced the importance of belief, making confirmation bias the strongest predictor of engagement. Recent studies suggest that inattention to accuracy increases fake news spreading [27, 79], so adding an AI recommendation may prompt greater inattention due to the influence exerted by AI.

Fourth, the effects of an AI recommendation were the same when we personified the AI agent by giving it a name and an image in Study 1, as is common [94], and when we presented it as an algorithm in Study 2, which is also common [112]. Study 1 was conducted in early 2022, before the explosion of generative AI, while Study 2 was conducted after it in late 2023 after generative AI. Users reacted to the AI recommendation similarly regardless of intervening years and the display format of AI (humanlike or agentic). We need more studies to investigate the impacts of AI and users' understanding of the role of AI, as well as ways to disclose the role of AI. Humanizing AI is a well proven method to improve acceptance of conversational AI, where the user and AI interact [20, 34]. When the task and information flow is unidirectional rather than interactive (as with AI recommendations), humanizing AI may have less influence. We need more research to understand the effects of making AI more humanlike on the recommendation of social media news articles and the boundary of anthropomorphism theory.

Finally, experts are prominent sources of social influence in news consumption. Users are influenced by expert recommendations because they have a lay belief that experts are knowledgeable, and capable, regardless of the fact that they have no idea of the real identity of the expert, and have no prior experience interacting with that specific person. Instead, perceptions of ability are conferred by the role held by the individual [105]. Future studies should examine the social influence of experts and whether the nature of the expert matters. The expert we displayed was a middle-aged white male. Research should investigate whether the effect is influenced by the expert's features, such as gender, age, race/ethnicity, and physical appearance.

Implication for Policy and Practice

This study provides important practical implications for government policymakers and social media platforms. The desire to disclose the role of AI in recommending news stories on social media has caused heated debates in many countries. Our study is an initial study on the effect of such disclosure. Unfortunately, labeling social media news stories as recommended by AI (whether a humanlike AI agent or embedded AI) was not as beneficial as policymakers expected. Such AI labels had asymmetric effects, reducing belief in true news but having no effect on fake news. Thus, governments and platforms should be very careful about proposing such measures for the transparency of AI usage. They are likely to *increase* the problem of fake news, not decrease it.

Additionally, firms increasingly invest to make their AI chatbots or other AI empowered service to be more humanlike, aiming to increase performance and drive business benefits. However, before devoting a large number of resources, managers should ensure that humanizing AI can indeed have positive effects on users within the desired context. In our context of social media news, embedded AI had similar effects as more humanlike AI.

Conclusion

In summary, social media users were *less* likely to believe true news articles that were labeled as recommended by AI, but the AI labels had no effect on fake news articles. Belief that a news article was true influenced the desire to share, but the presence of an AI recommendation label made belief that a story was true *less* important to the sharing decision; instead sharing was influenced more by confirmation bias for those articles labeled as recommended by AI. This study reveals that AI recommendations have an asymmetric social influence, and play an important role in users' news beliefs and engagement actions. Contrary to policymakers' hopes, labeling social media news articles as recommended by AI is not likely to help reduce the problem of fake news.

Note

1. See: <https://www.ncsl.org/technology-and-communication/legislation-related-to-artificial-intelligence>, http://www.cac.gov.cn/2022-01/04/c_1642894606364259.htm; <https://www.whitehouse.gov/ostp/ai-bill-of-rights>. See Online Supplemental Appendix 6 for more information.

</NOTES>

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No potential conflict of interest was reported by the author(s).

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